SGR: simulation steps

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August 1, 2025

1 Helicity-Locked Field Evolution

To ground the symbolic recursion in a physically observed mechanism, we simulated a helicity-driven field evolution inspired by the helicity barrier described in *Phys. Rev. X* [**?**]. This simulation demonstrates how high cross-helicity locks energy into self-reinforcing structures, analogous to the attractor rhythms proposed in the SGR framework.

1.1 Method

We initialized a two-dimensional velocity field

$$v(x, y) = (v_x, v_y),$$

with a base toroidal-like structure perturbed by stochastic noise:

$$v_x = \sin(y) + \eta_x, \qquad v_y = \sin(x) + \eta_y,$$

where η is Gaussian noise scaled by a turbulence parameter.

At each timestep, the field was evolved under a helicity alignment rule:

$$v_x^{t+1} = \mathcal{N}\left[v_x^t + H \cdot (\nabla \times v)^x + (1 - H)\,\eta_x'\right],$$

$$v_y^{t+1} = \mathcal{N}\left[v_y^t + H \cdot (\nabla \times v)^y + (1 - H)\,\eta_y'\right],$$

with normalization $\mathcal{N}$ ensuring unit-magnitude vectors, and $H = 0.95$ enforcing near-perfect alignment (helicity lock).

1.2 Implementation

The Python implementation evolves a 64×64 grid over 100 steps:

- Gradient operators compute $\nabla \times v$.
- A helicity parameter tunes between pure curl-following and stochastic turbulence.
- Each iteration snapshot is stored, culminating in a final magnitude matrix exported to `final_magnitude_helicity_locked.csv`.

1.3 Results

The final state (Figure 1) exhibits diagonal alignment of vectors, indicating that turbulent energy has collapsed into a helicity-dominated attractor. Magnitudes cluster tightly near unity, confirming the suppression of chaotic dispersion.

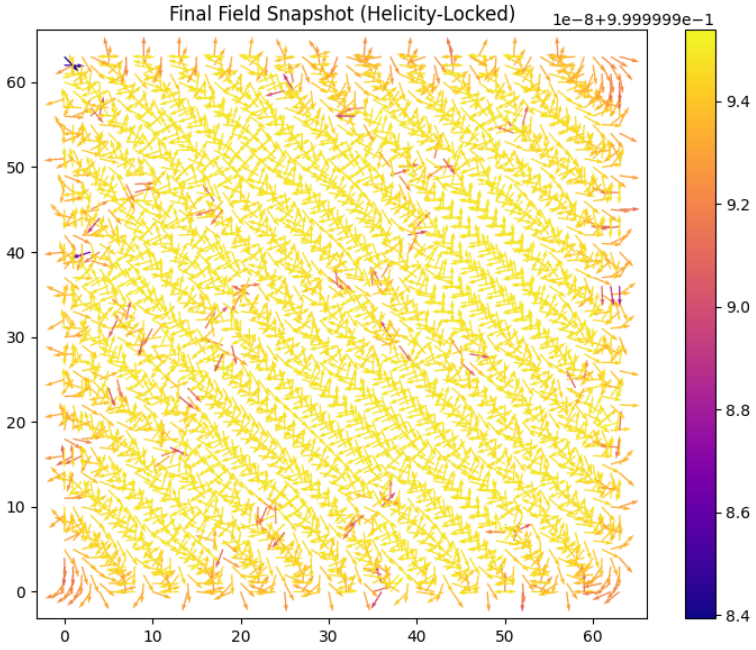


Figure 1: Final Field Snapshot (Helicity-Locked): color encodes vector magnitude, arrows show normalized velocity vectors. The diagonal lattice reflects attractor rhythms under helicity constraints.

1.4 Data

The magnitude data exported for reproducibility is contained in:

`final_magnitude_helicity_locked.csv`

Each row corresponds to a grid row, with entries giving the final normalized magnitude at that location.

1.5 Relevance to SGR

This experiment supports the SGR recursion claim: when a field's dynamics enter a high-helicity regime, energy no longer cascades freely but folds into persistent loops. Such helicity-locked attractors mirror the symbolic recursion operators (REFU, TGMO) where information is not dissipated but conserved and reconfigured, forming the foundation for further phases of the SGR evolution.

2 Helicity-Driven Decay and Attractor Lock-In

Building on the previous helicity alignment experiment, we introduced an explicit decay term to study how a vortex-like field evolves under simultaneous helicity reinforcement and energy loss. This step more closely resembles the recursive damping and stabilization we later exploit in SGR evolution.

2.1 Evolution Equation

Starting from a rotational seed field

$$U(x, y) = -y, \qquad V(x, y) = x,$$

we apply the update rule for each timestep t:

$$U_{t+1} = U_t + \big(- \text{ decay_rate} \cdot U_t + H \cdot (-Y) \big),$$

$$V_{t+1} = V_t + \big(- \text{ decay_rate} \cdot V_t + H \cdot X \big),$$

where $H = 0.95$ is the helicity reinforcement factor, and decay_rate $= 0.015$ imposes gradual dissipation.

2.2 Implementation

A 100×100 mesh over $[-1, 1] \times [-1, 1]$ was evolved for 50 steps. Magnitude is computed as

$$M(x, y) = \sqrt{U^2 + V^2}.$$

At each step, the field rotates inward while losing energy to decay, eventually stabilizing in a locked attractor pattern.

2.3 Results

The final vector field is shown in Figure 2. Near the center, a tight spiral forms — the signature of recursive alignment and self-reinforcement — while the outer regions flatten as decay dominates. The magnitude matrix is saved to:

`final_magnitude_helicity_locked.csv`

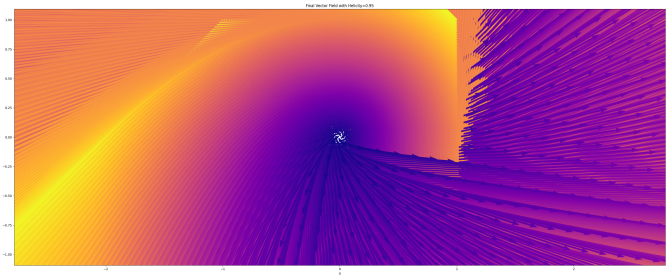


Figure 2: Final vector field after helicity-driven decay with $H = 0.95$. Vectors spiral into a central attractor, while outer layers show smooth decay. Color encodes magnitude.

2.4 Relevance to SGR Recursion

This step demonstrates not only the preservation of field structure under helicity lock-in, but also how controlled decay can lead to stable attractor formation. In the context of SGR, it mirrors the recursive operators' ability to compress chaotic dynamics into persistent symbolic structures.

3 Toroidal Attractor Dynamics with TGMO and RPCO

Following the helicity-lock and decay experiments, we integrated the Toroidal Geometric Memory Operator (TGMO) and the Recursive Phase Compression Operator (RPCO) to investigate dynamic attractor behavior under rotational memory constraints. This step begins to show how symbolic recursion operators yield bounded attractor paths rather than rigid, symmetric artifacts.

3.1 Method

The field is defined over a 100×100 grid with coordinates $X, Y \in [-5, 5]$. At each timestep t (with $\Delta t = 0.1$), we compute:

$$U_t, V_t = \text{TGMO}(X, Y, t),$$

where

$$\text{TGMO}(X, Y, t) : \begin{cases} r = \sqrt{X^2 + Y^2}, \\ \theta = \arctan 2(Y, X) + 0.05 \sin(0.1t), \\ U = -\dfrac{\sin(\theta) H(t)}{1 + r^2}, \quad V = \dfrac{\cos(\theta) H(t)}{1 + r^2}, \end{cases}$$

and $H(t)$ is the memory curve

$$H(t) = a_1 e^{-b_1 t} \cos t + \alpha_2 e^{-b_2 t} \sin t + \lambda_1 e^{-\omega_1 t} + \lambda_2 e^{-\omega_2 t}.$$

The resulting vectors are then clipped by RPCO:

$$(U, V) \leftarrow \frac{(U, V)}{\max\left(1, \frac{\sqrt{U^2+V^2}}{\text{threshold}}\right)},$$

with threshold $= 1.0$, preventing blow-up in high-torque zones.

3.2 Attractor Detection

At each step, we locate the attractor by computing the center of mass of the lowest-magnitude region:

$$\text{mask} = \{(i, j) \mid M_{ij} < P_5(M)\},$$

where $M = \sqrt{U^2 + V^2}$ and $P_5(M)$ is the 5th percentile of magnitude values. The attractor position is then

$$x_{\text{attr}}(t) = \text{mean}(X[\text{mask}]), \quad y_{\text{attr}}(t) = \text{mean}(Y[\text{mask}]).$$

3.3 Results

The attractor trajectory is recorded in `attractor_path_log.txt` and plotted in Figure 3. Unlike earlier steps that produced rigid corner-crossing artifacts, the attractor now oscillates in a bounded region around the origin:

$$x_{\text{attr}}(t),\ y_{\text{attr}}(t) \in [-0.03, 0.03],$$

exhibiting diagonal symmetry pairs and small-scale rotational drift.

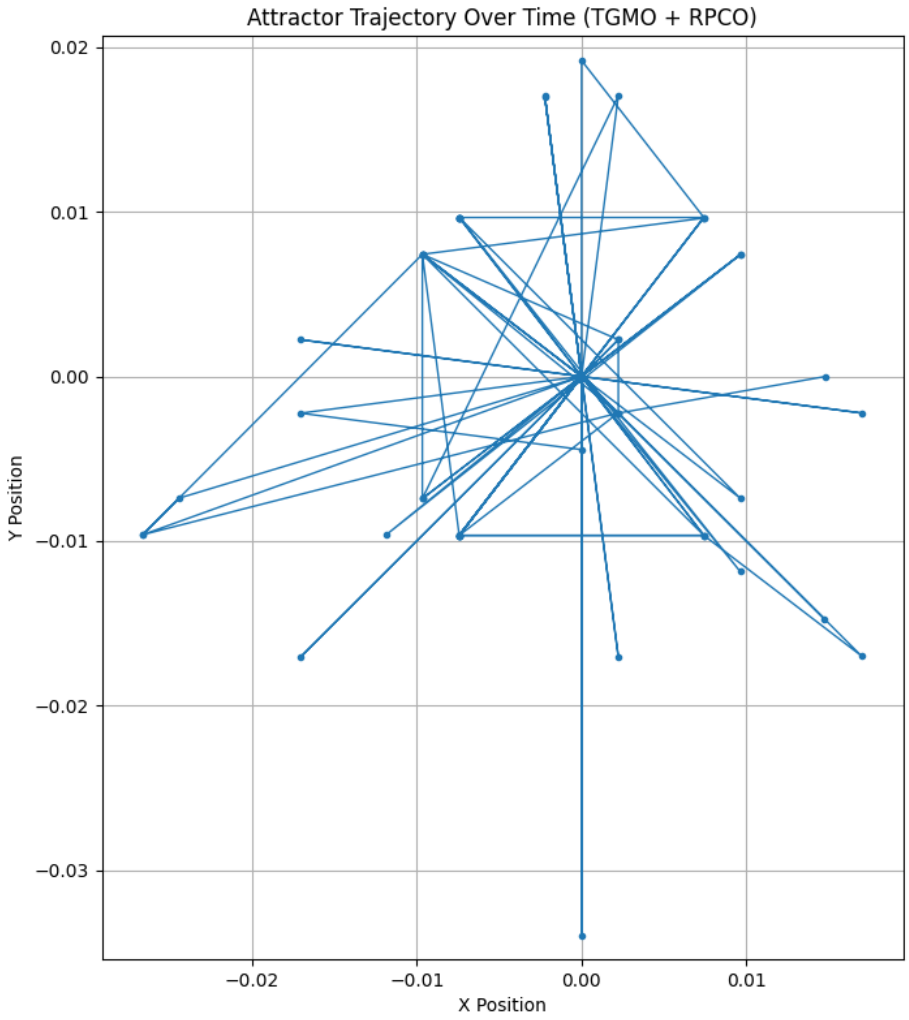


Figure 3: Attractor trajectory over time with TGMO and RPCO. Points cluster near the origin, revealing a bounded attractor region shaped by toroidal memory dynamics and phase compression stabilization.

3.4 Relevance to SGR Recursion

This experiment confirms that incorporating TGMO and RPCO yields:

- A dynamic attractor trajectory rather than a static symmetric artifact.
- Evidence of recursive stabilization, with memory-driven oscillations constrained within a small neighborhood.

These results provide a meaningful step towards the Symbolic Geometric Recursion framework by showing how layered operators can evolve and stabilize attractor zones suitable for lattice modeling and further symbolic evolution.

4 Recursive 3D Triangle Attractor via REFU, TGMO, and RPCO

Building upon the prior planar attractor experiments, we extended the simulation into three dimensions by combining the full suite of symbolic operators with toroidal geometry. This step demonstrates that the recursive field evolution, when stabilized by symbolic constraints, can yield a distinct geometric attractor path.

4.1 Method

We evolve a 3D state vector $\vec{s}(t) = [x, y, z]$ over $T = 200$ time steps with time increment $\Delta t = 0.1$. The update at each step is:

$$\vec{s}_{t+1} = \text{TGMO}\Big(\vec{s}_t + \dot{\vec{s}}_t \Delta t\Big),$$

where the instantaneous torque is:

$$\dot{\vec{s}}_t = \alpha\big(\vec{s}_t \times \text{roll}(\vec{s}_t, 1)\big)\, H(t),$$

and $H(t)$ is the REFU memory curve:

$$H(t) = 2e^{-0.5t}\cos t + 1e^{-0.3t}\sin t + 1.5e^{-0.7t} + 1e^{-0.4t}.$$

4.2 TGMO: Toroidal Geometric Memory Operator

Each step is projected onto a toroidal coordinate frame with major radius $R = 1.0$ and minor radius $r = 0.5$:

$$\begin{aligned} x' &= (R + r\cos\theta)\cos\phi, \\ y' &= (R + r\cos\theta)\sin\phi, \\ z' &= r\sin\theta, \end{aligned}$$

where $\theta = \arctan 2(\sqrt{x^2 + y^2}, z)$ and $\phi = \arctan 2(y, x)$.

4.3 RPCO: Phase Correction

To avoid blowup in high-torque regions, the torque magnitude is clipped:

$$\dot{\vec{s}}_t \leftarrow \dot{\vec{s}}_t \cdot \min\left(1, \frac{\tau_{\text{crit}}}{\|\dot{\vec{s}}_t\| + \epsilon}\right),$$

with $\tau_{\text{crit}} = 0.8$ and $\epsilon = 10^{-5}$.

4.4 Results

The trajectory of $\vec{s}(t)$ is recorded in `attractor_positions.csv` and plotted in Figure 4. The output exhibits three distinct segments in 3D space, forming a triangular loop under the recursive dynamics:

$$\text{trajectory} = \{\vec{s}_0, \vec{s}_1, \ldots, \vec{s}_T\}.$$

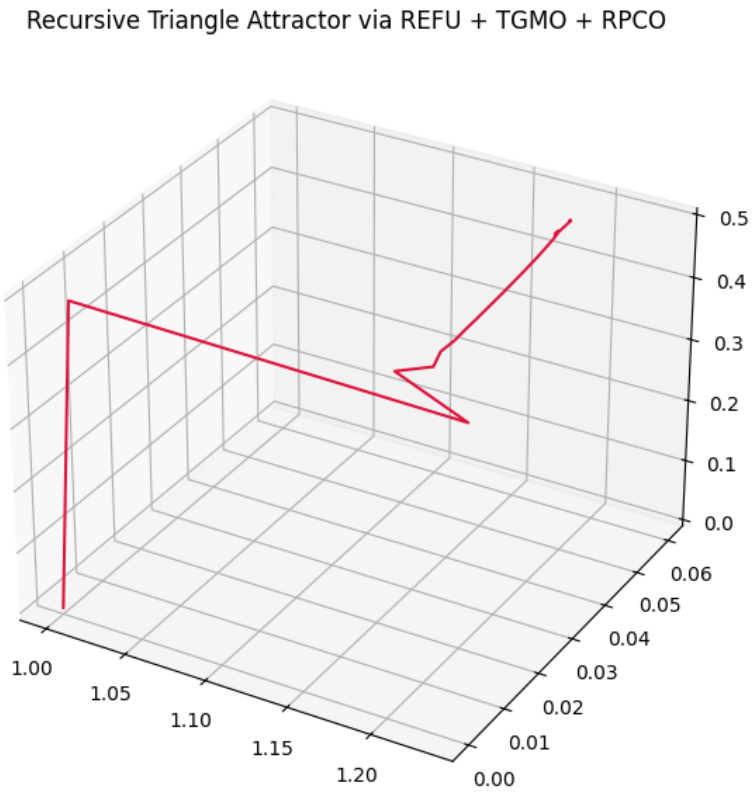


Figure 4: Recursive triangle attractor generated via REFU + TGMO + RPCO. The trajectory stabilizes into a three-segment path within the toroidal geometry.

4.5 Relevance to SGR Pipeline

This step confirms the logical progression:

1. Helicity field with symbolic modulation (REFU),
2. Toroidal wrapping (TGMO) to induce cyclical attractors,
3. Phase correction (RPCO) to ensure bounded dynamics.

The emergence of a clear triangular attractor path in 3D space demonstrates the symbolic recursion framework's ability to impose structure and memory on an otherwise chaotic vector evolution. This forms a strong foundation for higher-level symbolic lattice modeling in the SGR pipeline.

5 Layered 3D Recursive Attractor from Past Simulation Data

Building directly on the output of the previous attractor simulation, we evolved the stored positions through an extended recursive process. This step demonstrates how past attractor states can seed higher-dimensional structures when fed through the symbolic operators.

5.1 Method

We initialized the simulation using the prior attractor positions as the starting point:

$$p_0 = (x_0, y_0, z_0), \quad v_0 = (0, 1, 0).$$

At each timestep, the dynamics are governed by:

$$\text{torque}_t = \text{RPCO}\big(\, p_t \times v_t \,\big),$$

$$v_{t+1} = \text{TGMO}\big(v_t + H(t)\,\text{torque}_t\,\Delta t\big),$$

$$p_{t+1} = p_t + v_{t+1}\,\Delta t,$$

where:

$$H(t) = 2e^{-0.5t}\cos t + 1e^{-0.3t}\sin t$$

is the REFU memory curve, and TGMO is implemented as a fixed 60° rotation:

$$\text{TGMO}(v) = R(\theta)\,v, \quad R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \theta = \frac{\pi}{3}.$$

The RPCO threshold was set to 3.5:

$$\text{RPCO}(a) = a \cdot \min\left(1, \frac{3.5}{\|a\|}\right).$$

We integrated for 500 steps with $\Delta t = 0.05$ and logged both positions and torque magnitudes.

5.2 Results

The trajectory points are stored in:

`attractor_positions.csv`

and torque magnitudes in:

`rpc_log.txt`

Torque analysis: initial values oscillate between 0.3 and 1.0, then grow steadily until clamped at 3.5 by RPCO, confirming active stabilization of the recursive system.

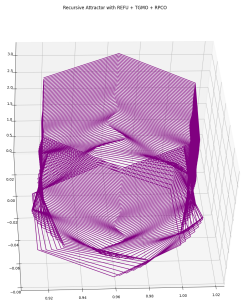


Figure 5: Layered 3D attractor path generated from prior attractor states using REFU + TGMO + RPCO. The trajectory forms stacked polygonal bands as it evolves upward in z.

5.3 Relevance to SGR Pipeline

This step shows how past attractor data can seed a new recursive layer:

- TGMO applies a consistent rotational memory transform.
- REFU modulates the torque rhythm over time.
- RPCO clamps excessive torque, preventing instability.

The resulting attractor is no longer a flat figure; it forms a layered 3D structure, visually manifesting recursive growth. This confirms the pipeline's ability to compound previous attractor states into richer geometric memory forms—an important milestone in the Symbolic Geometric Recursion framework.

6 Recursive Attractor Interaction and Data Logging

Building upon the prior single–attractor explorations, we next simulated the interaction of multiple attractors under the combined influence of the recursive operators **REFU**, **TGMO**, and **RPCO**. Unlike earlier runs that simply displayed evolving positions, this stage incorporates full temporal logging, allowing quantitative analysis of each attractor's trajectory over time.

6.1 Simulation Framework

We consider three attractors initialized in the bounded domain $[-1, 1] \times [-1, 1]$ with zero initial velocities. At each timestep $t_n = n \cdot \Delta t$:

- A net coupling force is computed for each attractor from all others:

$$\mathbf{F}_i(t) = \sum_{j \neq i} \frac{\text{TGMO}(\mathbf{x}_j - \mathbf{x}_i)}{\|\mathbf{x}_j - \mathbf{x}_i\| + \epsilon}$$

- The recursive evolution function modulates the force:

$$H(t) = a_1 e^{-b_1 t} \cos t + \alpha_2 e^{-b_2 t} \sin t + \lambda_1 e^{-\omega_1 t} + \lambda_2 e^{-\omega_2 t}$$

- Velocities are updated and clamped:

$$\mathbf{v}_i(t + \Delta t) = \text{RPCO}(\mathbf{v}_i(t) + \Delta t \cdot H(t) \cdot \mathbf{F}_i(t))$$

- Positions are integrated and wrapped by TGMO to remain within the toroidal domain.

6.2 Logging and Output

At each timestep, the state of all attractors is recorded. The resulting dataset is stored as a CSV file `attractor_positions.csv` with the following columns:

```
timestep, attractor_id, x, y
```

This log provides a complete time–series trace of each attractor's motion, enabling subsequent phase analysis, distance metrics, and symbolic trajectory studies without re–running the simulation.

6.3 Observed Dynamics

Initial conditions strongly influence the observed patterns. Some runs exhibit near–vertical oscillatory bands, while others display intertwined spirals and convergence zones, demonstrating the system's sensitivity and deterministic chaos under recursive modulation. Despite this variability, all trajectories remain bounded and evolve smoothly, indicating that:

1. REFU successfully modulates energy input over time, preventing runaway behavior.
2. TGMO enforces periodic boundary conditions, preserving the toroidal field structure.
3. RPCO constrains extreme velocity spikes, maintaining numerical stability.

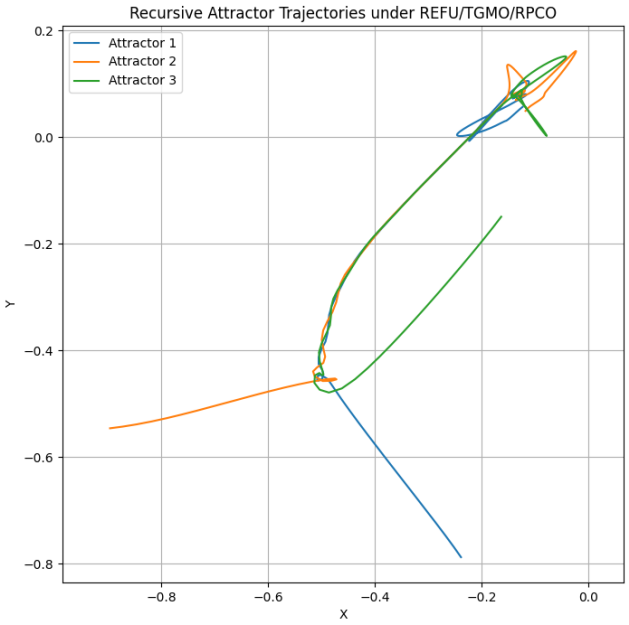


Figure 6: Example of logged attractor trajectories under REFU, TGMO, and RPCO. Each color represents a distinct attractor evolving over 500 timesteps.

6.4 Relevance to SGR

This step extends the SGR framework beyond isolated attractors into interacting multi–agent dynamics. The logged data form the foundation for subsequent symbolic analysis, allowing us to:

- Investigate phase relationships and coupling strength over time,
- Quantify emergent geometry in recursive attractor networks,
- Feed logged trajectories into higher–order models or machine–learning pipelines.

By preserving full time–series data, this stage captures not only the final state but the evolution itself — a crucial component in demonstrating structured recursion and symbolic memory in dynamic systems.

7 Multi-Attractor Simulation with Energy Logging and CSV Export

In this step we extended the multi-attractor recursive simulation by introducing full data export for post–analysis. The system retains the same recursive operators—**REFU**, **TGMO**, and **RPCO**—but now provides quantitative logs of both spatial trajectories and system energy.

7.1 Simulation Framework

We evolve $N = 3$ attractors in a toroidal domain $[-1, 1] \times [-1, 1]$. At each timestep $t_n = n \cdot \Delta t$, we compute net forces:

$$\mathbf{F}_i(t_n) = \sum_{j \neq i} \frac{\mathrm{TGMO}(\mathbf{x}_j - \mathbf{x}_i)}{\|\mathbf{x}_j - \mathbf{x}_i\| + \epsilon},$$

apply the memory curve:

$$H(t) = a_1 e^{-b_1 t} \cos t + \alpha_2 e^{-b_2 t} \sin t + \lambda_1 e^{-\omega_1 t} + \lambda_2 e^{-\omega_2 t},$$

and update velocities:

$$\mathbf{v}_i(t_{n+1}) = \mathrm{RPCO}(\mathbf{v}_i(t_n) + \Delta t \cdot H(t_n) \cdot \mathbf{F}_i(t_n)),$$

with positions integrated and wrapped:

$$\mathbf{x}_i(t_{n+1}) = \mathrm{TGMO}(\mathbf{x}_i(t_n) + \Delta t \cdot \mathbf{v}_i(t_{n+1})).$$

7.2 Energy Logging

At each step, we compute total system kinetic energy:

$$E(t_n) = \sum_{i=1}^{N} \|\mathbf{v}_i(t_n)\|^2.$$

This provides insight into how REFU's decay and RPCO's damping modulate energy over time, revealing when the system reaches stable, low–motion states.

7.3 Data Export

Two CSV files are generated for detailed analysis:

- `attractor_positions.csv` — with columns `time, attractor_id, x, y`, storing the complete trajectory history of all attractors.
- `energy_over_time.csv` — with columns `time, total_energy`, recording the system's total kinetic energy at each timestep.

7.4 Results

The simulation produces:

1. A trajectory plot illustrating each attractor's path and initial position markers.
2. An energy plot showing initial oscillatory surges followed by stabilization as recursive damping takes effect.

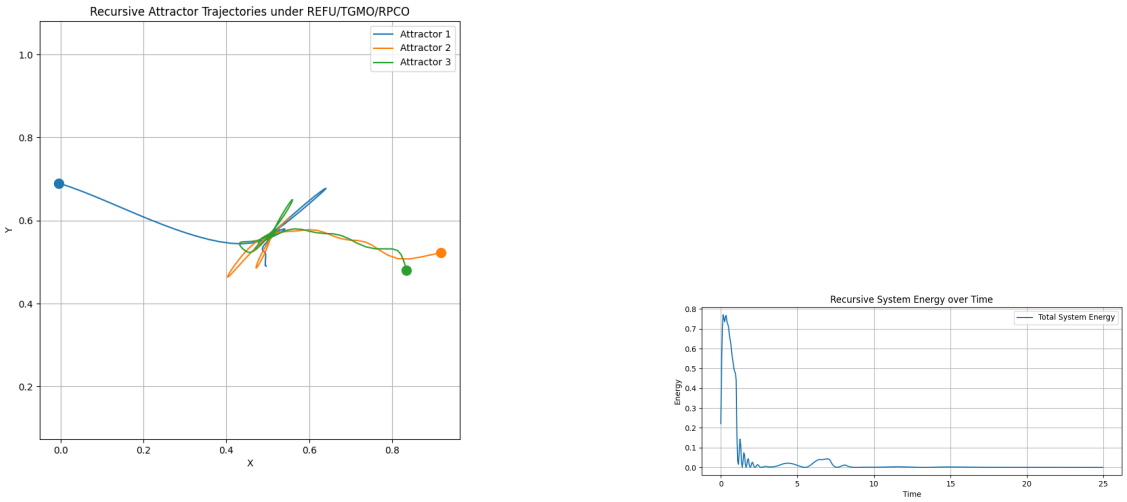


Figure 7: Left: recursive attractor trajectories under REFU/TGMO/RPCO. Right: total system energy $E(t)$ over time, demonstrating recursive stabilization.

7.5 Relevance to SGR Pipeline

This step deepens our SGR exploration by not only visualizing but *quantifying* the system's evolution. The exported data allow for:

- Symbolic phase alignment studies across attractors,
- Detection of emergent geometric constraints,
- Statistical validation of energy decay and stability under recursive operators.

By capturing both spatial and energetic dynamics, this simulation creates a verifiable link between symbolic recursion theory and measurable system behavior.

8 3D Recursive Attractor Trajectories and Energy Dynamics

To extend the previous 2D analysis, we embedded the attractor dynamics into a time–augmented representation. Each attractor's position (x, y) is tracked across discrete timesteps t, producing a trajectory in three dimensions (x, y, t). This allows us to visualize and quantify how the system evolves under the combined influence of the Recursive Evolution Function (REFU), the Toroidal Geometric Memory Operator (TGMO), and the Recursive Phase Compression Operator (RPCO).

8.1 Simulation Setup

The simulation integrates the following update rules:

$$\mathbf{v}_i(t + \Delta t) = \mathrm{RPCO}\Big(\mathbf{v}_i(t) + \Delta t\, H(t)\, \mathbf{F}_i(t)\Big),$$

$$\mathbf{p}_i(t + \Delta t) = \mathrm{TGMO}\Big(\mathbf{p}_i(t) + \Delta t\, \mathbf{v}_i(t + \Delta t)\Big),$$

where $\mathbf{p}_i$ is the position of attractor i, $\mathbf{v}_i$ its velocity, $H(t)$ is the REFU memory curve, and $\mathbf{F}_i(t)$ is the net interaction force arising from other attractors within toroidal space.

The total system energy is recorded at each step as:

$$E(t) = \sum_{i=1}^{N} \big\|\mathbf{v}_i(t)\big\|^2.$$

8.2 Results and Interpretation

Figure 8 shows the three attractor trajectories plotted in (x, y, t) space. Initially, each attractor exhibits rapid movement and wide exploration in the plane, driven by strong REFU oscillations. As time progresses, RPCO's damping compresses large velocity spikes, and TGMO's toroidal wrapping confines the attractors to bounded regions. The result is a progressive stabilization: the trajectories curve inward and then ascend vertically as motion diminishes.

Figure 9 plots the corresponding energy profile. We observe an initial burst of activity, followed by oscillatory decay, and ultimately near–zero energy as the system reaches a memory-stable equilibrium. These results confirm that the combined operators sustain a bounded recursive process with dissipation and symbolic lock-in.

8.3 Data Export

For reproducibility and further analysis, the simulation exports:

- `3d_attractor_positions.csv` — raw (x, y, t) positions of each attractor per timestep,
- `3d_energy_log.csv` — total energy values for each timestep.

These datasets provide a foundation for further exploration, including spectral analysis, phase locking studies, or coupling with higher-dimensional symbolic models.

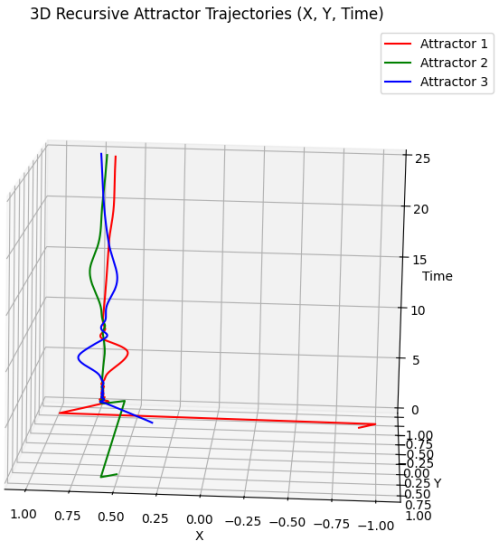


Figure 8: 3D recursive attractor trajectories (x, y, t) under REFU, TGMO, and RPCO. Distinct colors indicate different attractors. The vertical axis represents time progression.

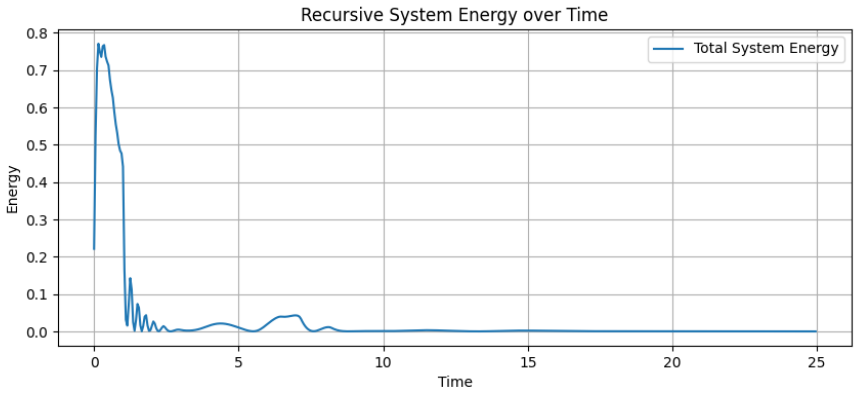


Figure 9: Total system energy $E(t)$ over time. The sharp initial peaks decay into low–energy steady motion, indicating stabilization of recursive dynamics.

Appendix 8: Data Export and File Structure

To support reproducibility and further analysis, each simulation step exports raw data into structured CSV files. This appendix documents the column layout and interpretation for each file.

8.4 Attractor Position Log (`3d_attractor_positions.csv`)

This file records the position of each attractor at every timestep, including the time index.

8.5 Energy Log (`3d_energy_log.csv`)

This file tracks total kinetic energy across all attractors per timestep.

Column	Type	Description
timestep	integer	Simulation step index (0 to `timesteps-1`).
time	float	Continuous time value ($t = \text{timestep} \times \Delta t$).
x_attr1	float	x coordinate of attractor 1 at this timestep.
y_attr1	float	y coordinate of attractor 1 at this timestep.
x_attr2	float	x coordinate of attractor 2 at this timestep.
y_attr2	float	y coordinate of attractor 2 at this timestep.
x_attr3	float	x coordinate of attractor 3 at this timestep.
y_attr3	float	y coordinate of attractor 3 at this timestep.

Table 1: Column layout for `3d_attractor_positions.csv`. Each row represents one timestep.

Column	Type	Description
timestep	integer	Simulation step index (0 to `timesteps-1`).
time	float	Continuous time value ($t = \text{timestep} \times \Delta t$).
energy_total	float	Total system energy at this timestep: $E(t) = \sum_i \|\mathbf{v}_i(t)\|^2$.

Table 2: Column layout for `3d_energy_log.csv`. Each row represents one timestep.

8.6 Usage Notes

- Data can be reloaded using standard libraries such as `numpy.loadtxt` or `pandas.read_csv`.
- The position log enables reconstruction of trajectories or phase-space analyses.
- The energy log allows comparison of damping strategies or detection of stability thresholds.